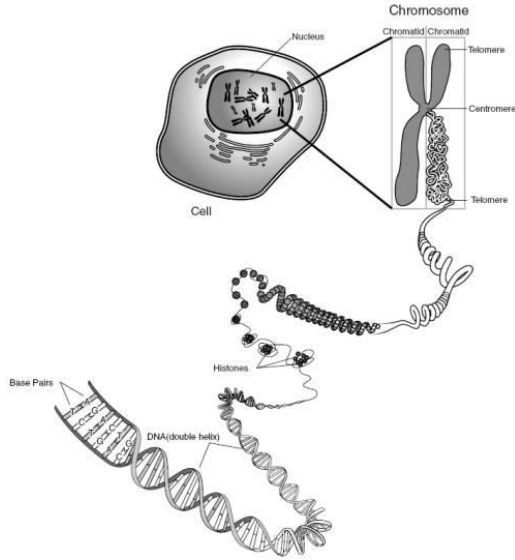
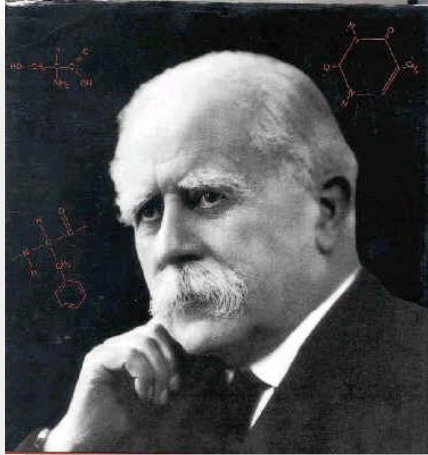




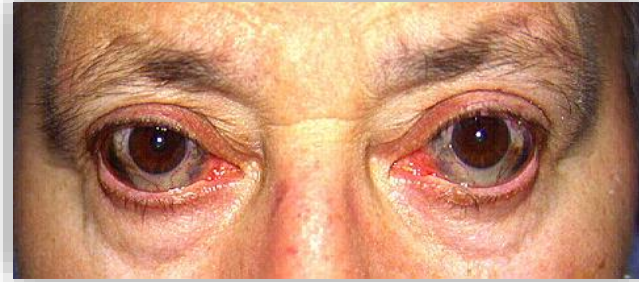
Què son els gens? Quina és la naturalesa química dels gens? Genètica molecular i metabolisme



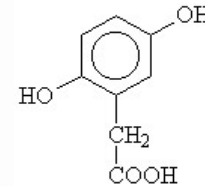
Archibald Garrod i els errors congènits del metabolisme



Alcaptonúria

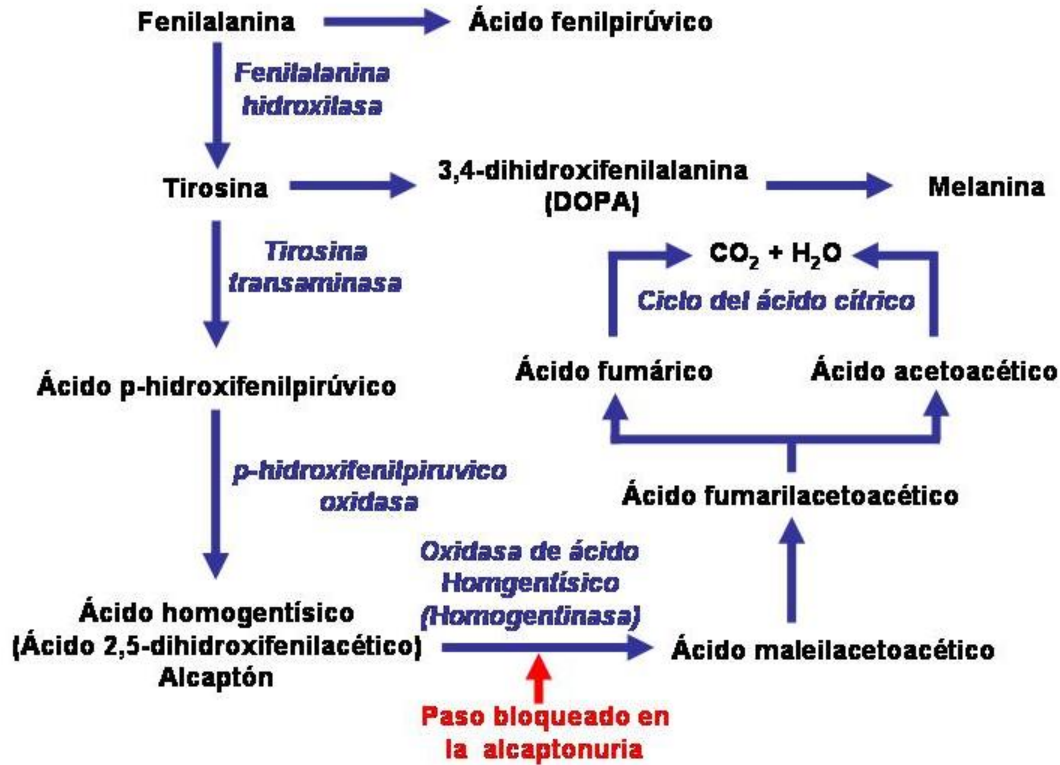


Taques fosques en ulls i orelles



Àcid homogentísic



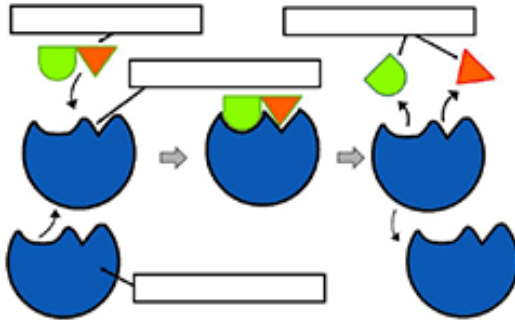


Els enzims, què cal saber?

Name _____ Date _____

Analyzing Graphics: Enzymes

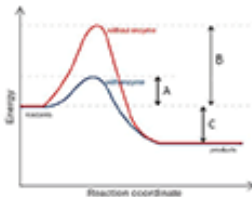
1. Enzymes act on substrates. Label the enzyme, substrate, active site, and products on diagram.



Answer true or false to the following statements based on the graphic:

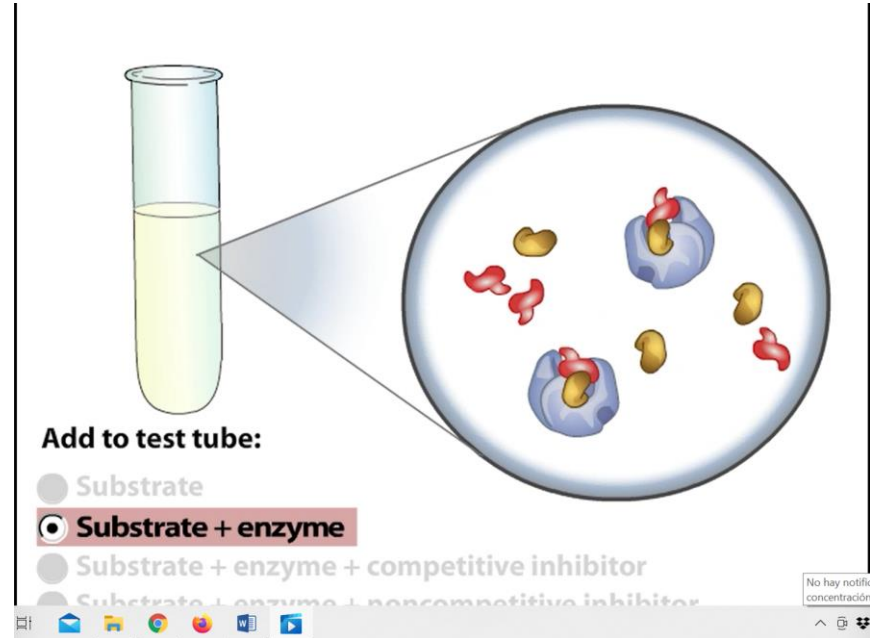
- a. _____ Enzymes interact with many different substrates.
- b. _____ Enzymes change shape after a reaction occurs.
- c. _____ An enzyme can be reused with a new substrate.
- d. _____ The substrate is changed in the reaction.
- e. _____ If the shape of the enzyme changed, it would no longer work.
- f. _____ When all substrates are used, the reaction stops.

2. Enzymes speed up the reaction by lowering the activation energy needed for the reaction to start. Compare the activation energy with and without the enzyme.



Identify the part of the graph that shows:

- a) ___ Overall energy released during reaction
b) ___ Activation energy with enzyme
c) ___ Activation energy without enzyme



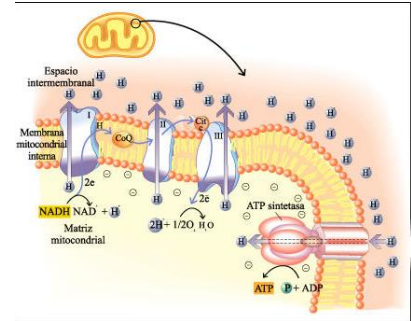
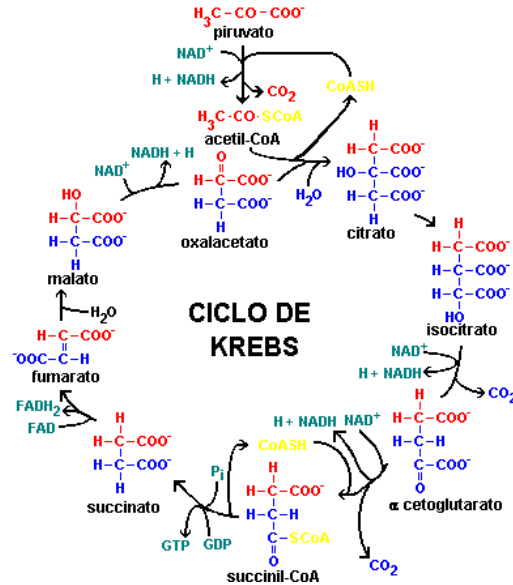
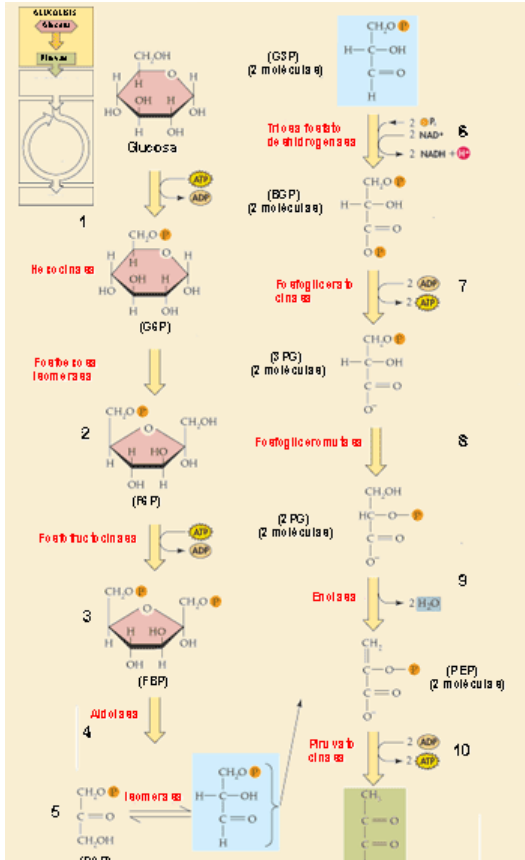
http://bcs.whfreeman.com/WebPub/Biology/hillis1e/Animated%20Tutorials/at0302/at_0302_enzyme_catalysis.html



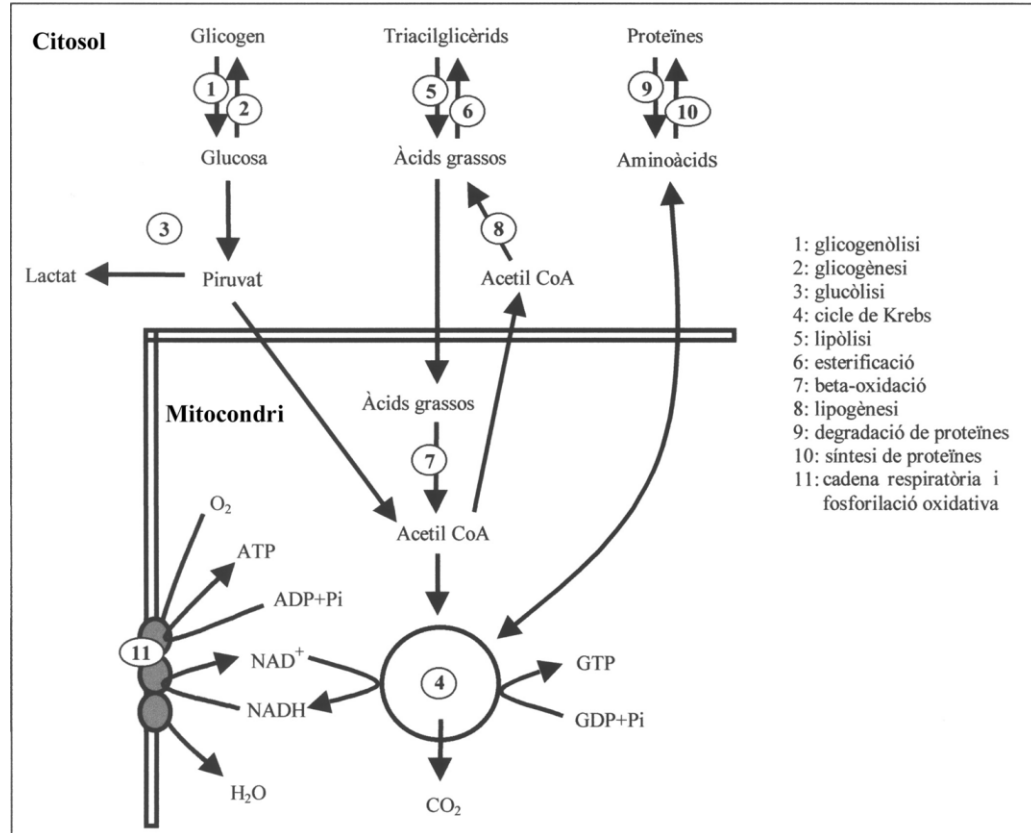
Glucosa + 6 O₂



6H₂O + 6CO₂



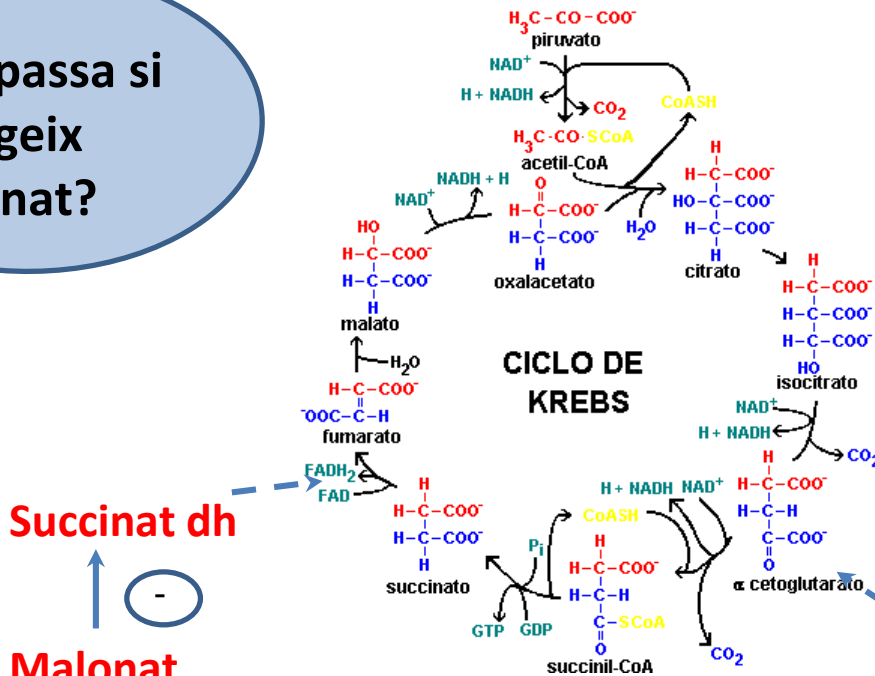
ESQUEMA GENERAL DEL METABOLISME



Quin nivell a batxillerat?

Què passa si
s'afegeix
malonat?

Què passa si hi ha
una deficiència de
vitamina B1?



Cetoglutàric dh

**Funciona amb el coenzim tiamina
pirofosfat (TPP –vitamina B1-)**



Com s'arriba al coneixement de les vies metabòliques?

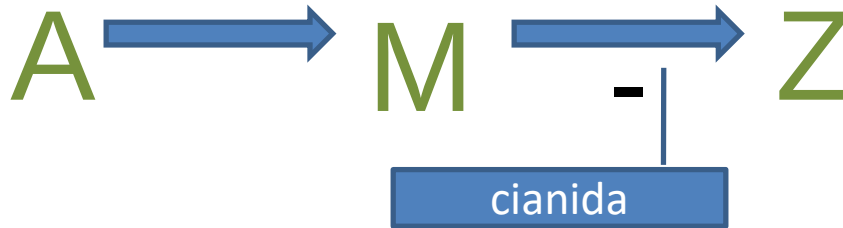


Un verí
enzimàtic

+ cianida:



Què es pot deduir?



**Però, continuem sense saber què son els
gens.**

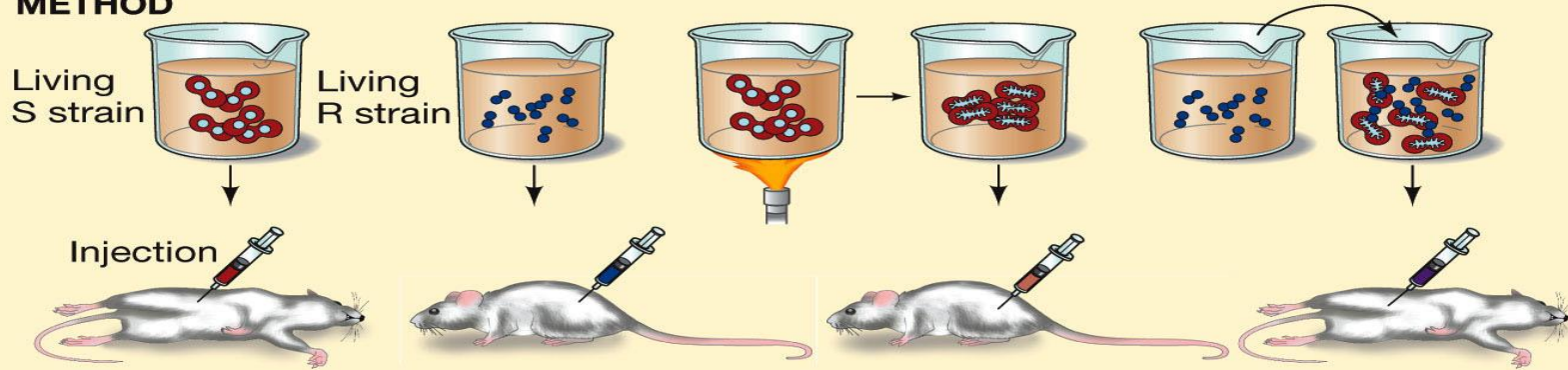
**Dos experiments clau per a la
determinació de la naturalesa del
material genètic:**



EXPERIMENT

HYPOTHESIS: Material in dead bacterial cells can genetically transform living bacterial cells.

METHOD



RESULTS

Mouse dies
Living S strain
cells found in
heart

Mouse healthy
No bacterial cells
found in heart

Mouse healthy
No bacterial cells
found in heart

Mouse dies
Living S strain
cells found in
heart

CONCLUSION: A chemical substance from one cell is capable of genetically transforming another cell.

LIFE 8e, Figure 11.1

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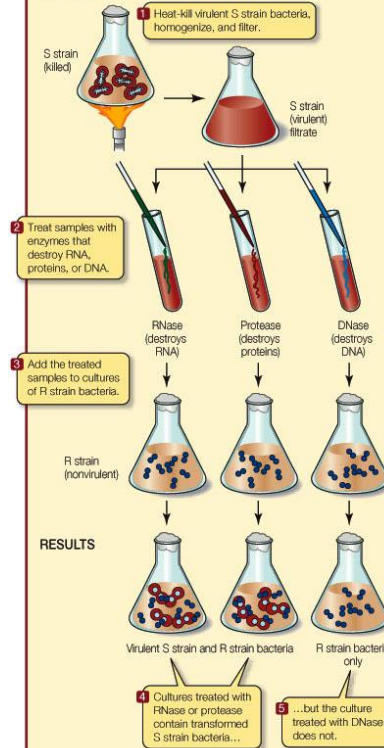
1928, Frederick Griffith: una substància química provinent d'una cèl·lula és capaç de transformar genèticament una altra cèl·lula



EXPERIMENT

HYPOTHESIS: The chemical nature of the transforming substance from pneumococcus is DNA.

METHOD



CONCLUSION: Because only DNase destroyed the transforming substance, the transforming substance is DNA.

1944 Avery, McLeod i
McCarty: La
substància
transformant és el
DNA

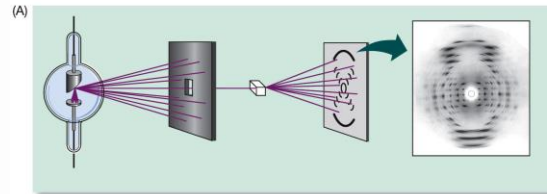
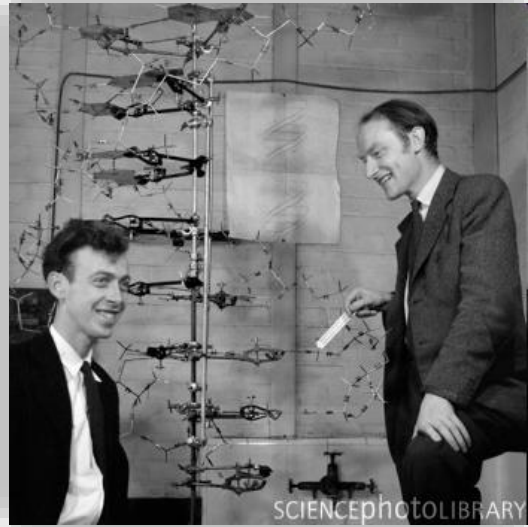


Si el DNA és el material hereditari, llavors...

- les cèl·lules han de poder duplicar el DNA (és a dir fer còpies exactes)
- ha de ser capaç de portar informació genètica
- ha de poder traslladar la informació fins el lloc de la cèl·lula que realitza la funció
- el DNA ha de poder mutar

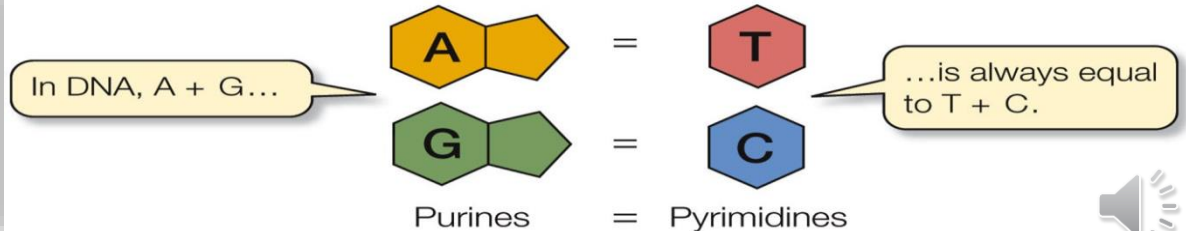


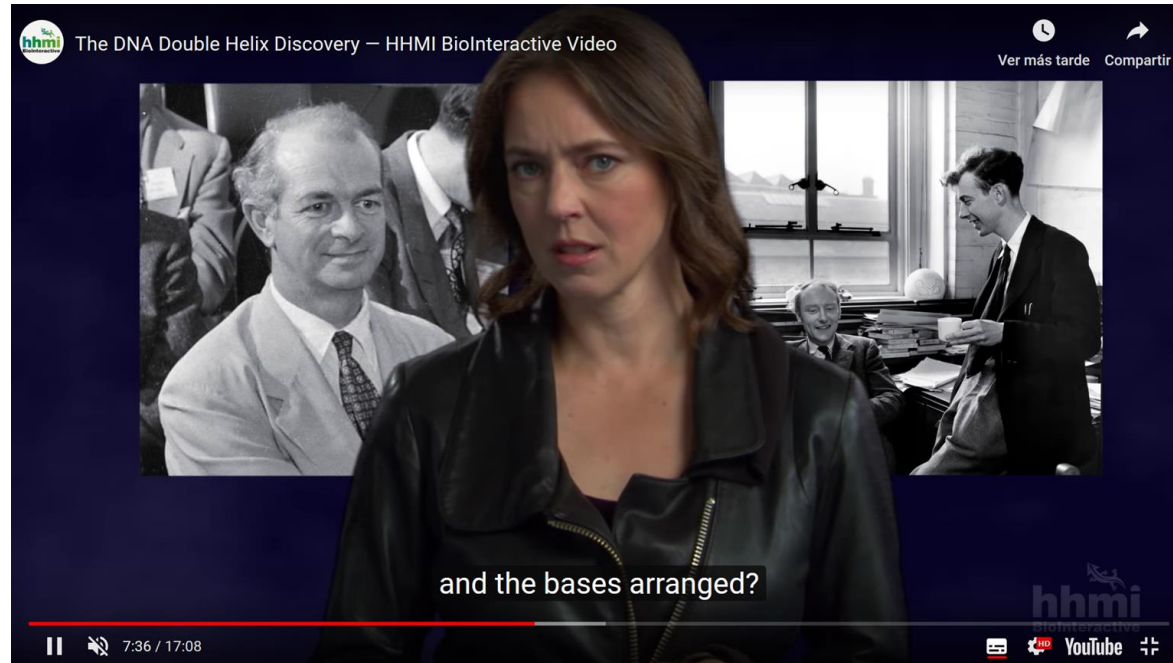
1953, Watson i Crick: el model de DNA



LIFE 8e, Figure 11.6

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Un bon recurs: The DNA Double Helix Discovery

<https://www.hhmi.org/biointeractive/double-helix>

https://www.youtube.com/watch?time_continue=886&v=1vm3od_UmFg



Com treballar amb l'alumnat?



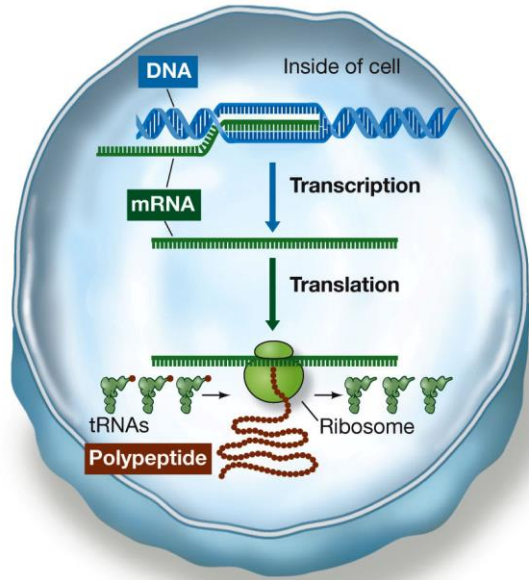
Modeling the Structure of DNA

<https://www.biointeractive.org/classroom-resources/modeling-structure-dna>

How DNA Is Packaged

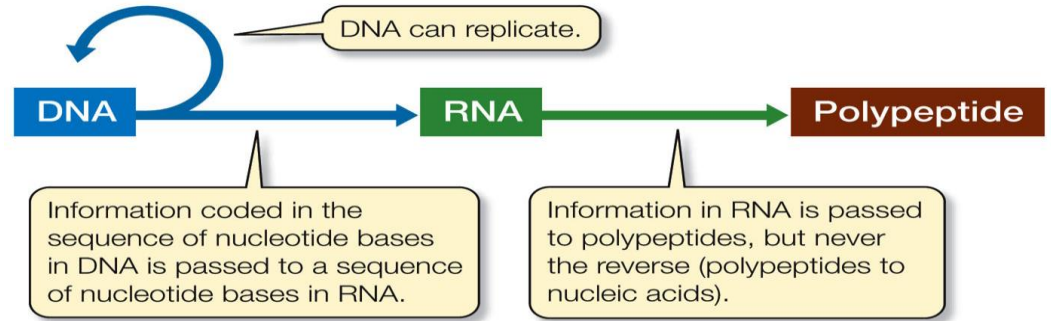
<https://www.biointeractive.org/classroom-resources/how-dna-packaged>





LIFE 8e, Figure 12.3

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LIFE 8e, Figure 12.2

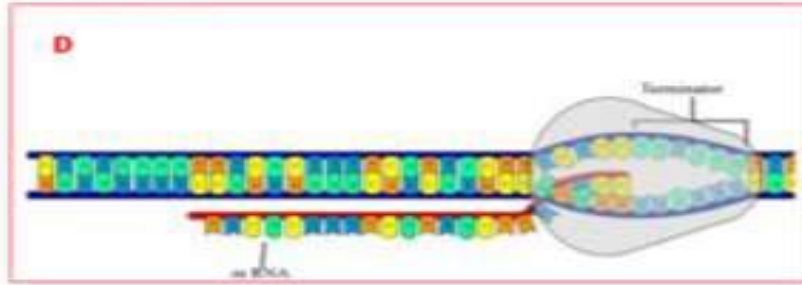
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Super-resum del dogma central:

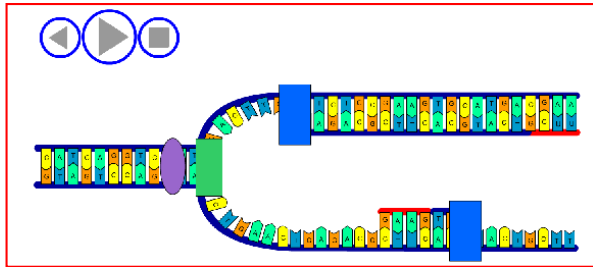
<https://media.hhmi.org/biointeractive/click/genetic-medicine-interactive/>



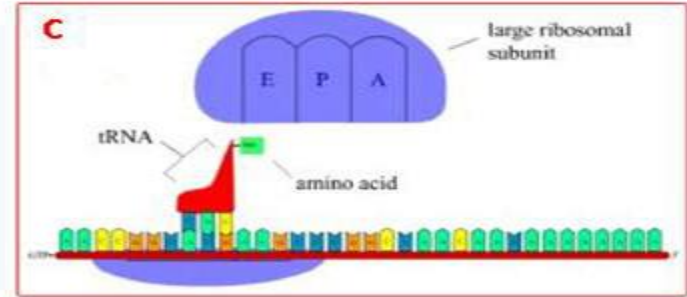
Com treballar amb l'alumnat?



<http://www.stolaf.edu/people/giannini/flashanimat/molgenetics/transcription.swf>



<http://www.stolaf.edu/people/giannini/flashanimat/molgenetics/dna-rna2.swf>



<http://www.stolaf.edu/people/giannini/flashanimat/molgenetics/translation.swf>



<http://ncbi.nlm.nih.gov/genome/guide/human>



<https://www.youtube.com/user/ndsuvirtualcell>



Per repassar conceptes bàsics:
<https://www.khanacademy.org/>